

6. A prime number is only divisible by one and itself. The first five prime numbers are shown below:

2, 3, 5, 7, 11, ...

The code below checks if a number is prime.

```

Line 1  FUNCTION checkPrime(INTEGER n) RETURNS BOOLEAN
Line 2      DECLARE validPrime INITIALLY TRUE
Line 3      IF n < 2 THEN
Line 4          SET validPrime TO FALSE
Line 5      ELSE
Line 6          FOR divisor FROM 2 TO (n-1) DO
Line 7              IF <the remainder of n divided by divisor is
                  equal to 0> THEN
Line 8                  SET validPrime TO FALSE
Line 9              END IF
Line 10         END FOR
Line 11     RETURN validPrime
Line 12 END FUNCTION
...
Line 42 DECLARE inputNum AS INTEGER INITIALLY FROM KEYBOARD
Line 43 <set isPrime by calling the function checkPrime to
        identify if inputNum is prime or not>
Line 44 IF isPrime = TRUE THEN
Line 45     SEND inputNum & " is prime." TO DISPLAY
Line 46 ELSE
Line 47     SEND inputNum & " is not prime." TO DISPLAY
Line 48 END IF

```

- (a) Using a programming language of your choice, write the code for line 7.

2

- (b) Using a programming language of your choice, write the code for line 43.

2